

Homework 6

PROBLEM 5.47

A linear-phase FIR system has a real impulse response $h[n]$ whose z -transform is known to have the form

$$H(z) = (1 - az^{-1})(1 - e^{j\pi/2}z^{-1})(1 - bz^{-1})(1 - 0.5z^{-1})(1 - cz^{-1})$$

where a, b, c are zeros of $H(z)$. Also, we know that $H(e^{j0}) = 0$.

- (a) $H(z)$ is a product of five first-order factors, so $H(z)$ is a 5th-degree polynomial in z^{-1} . Therefore the impulse response has length $M + 1 = 5 + 1 = 6$ samples.
- (b) For a real, linear-phase FIR system, if the length is 6 then $M = 5$ is odd. The problem also tells us $H(e^{j0}) = 0$, i.e. there is a zero at $z = 1$. A real, linear-phase FIR of odd order with $H(e^{j0}) = 0$ corresponds to a Type IV system. So this is Type IV.
- (d) Because $H(e^{j0}) = 0$, there must be a zero at $z = 1$, so set $a = 1$.

There is a given zero on the unit circle at $z = e^{j\pi/2}$. For linear phase and real $h[n]$, the conjugate zero must also be present, so there must be a zero at $z = e^{-j\pi/2}$. That zero must come from the unknown factor $(1 - bz^{-1})$, so set $b = e^{-j\pi/2}$.

Finally, there is a zero at $z = 0.5$. For linear phase, zeros off the unit circle must occur in reciprocal pairs, so we must also have a zero at $z = 2$. That must be the remaining factor, so set $c = 2$.

Thus

$$a = 1, \quad b = e^{-j\pi/2}, \quad c = 2.$$

- (e) With the zeros found above,

$$H(z) = (1 - z^{-1})(1 - e^{j\pi/2}z^{-1})(1 - e^{-j\pi/2}z^{-1})(1 - 0.5z^{-1})(1 - 2z^{-1}).$$

Multiplying these gives

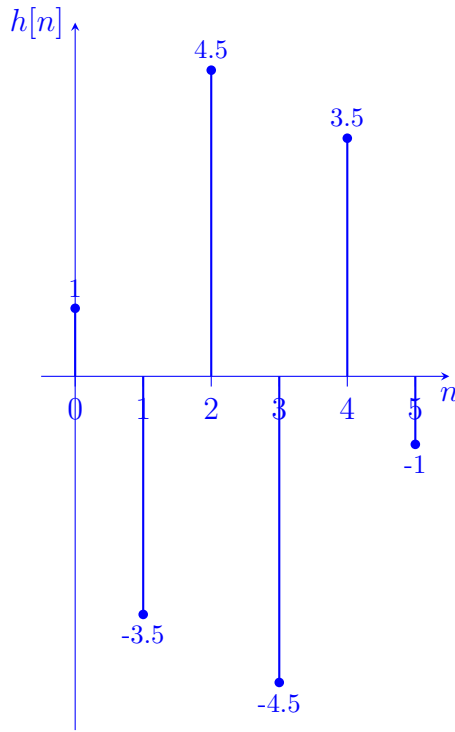
$$H(z) = 1 - 3.5z^{-1} + 4.5z^{-2} - 4.5z^{-3} + 3.5z^{-4} - z^{-5}.$$

So the impulse response is

$$h[0] = 1, \quad h[1] = -3.5, \quad h[2] = 4.5, \quad h[3] = -4.5, \quad h[4] = 3.5, \quad h[5] = -1.$$

This is consistent with a Type IV symmetry $h[n] = -h[5 - n]$.

Here is the stem plot of the impulse response...



PROBLEM 5.55

Each of the pole-zero plots in ith the specification of the ROC, describes an LTI system with system function $H(z)$. Determine whether any of the following statements are true. Justify your answer with a brief statement or a counterexample.

- (a) The system is a zero-phase or a generalized linear-phase system.
- (b) The system has a stable inverse $H_i(z)$.

(1) a zero-phase or generalized linear-phase system must have all poles and zeros in conjugate reciprocal pairs (with the allowed exceptions at $z = 0, \infty, \pm 1$); (2) a stable system has an ROC that includes the unit circle.

Case (a):

There are poles at $z = \frac{1}{2}$ (on the positive real axis) and also at two symmetric locations inside the unit circle, but they are not arranged in conjugate-reciprocal pairs. Hence it is not zero-phase or generalized linear-phase.

For the inverse: since $H(z)$ has a zero at the origin, $H_i(z)$ will have a pole at $z = 0$. The given ROC for $H(z)$ is $|z| < \frac{1}{2}$, so for $H_i(z)$ the ROC is $0 < |z| < \infty$, which contains the unit circle, so the inverse is stable (and if the ROC of $H(z)$ also included $z = \infty$, the inverse would be causal as well).

So, for (a): false. For (b): true.

Case (b):

The poles are not in conjugate-reciprocal pairs, so it is not zero-phase or generalized linear-phase.

For the inverse: $H(z)$ already has poles outside the unit circle and ROC $|z| > \frac{3}{2}$. The inverse would have poles at the zeros of $H(z)$, which lie strictly inside the unit circle, so the inverse ROC must include the unit circle to be stable. Here that ROC is $|z| \geq \frac{3}{2}$ for $H(z)$, so for $H_i(z)$ the ROC is $|z| > \frac{3}{2}$, which includes the unit circle. Thus the inverse is stable and causal.

So, for (a): false. For (b): true.

Case (c):

The zeros occur in conjugate-reciprocal pairs (one at $z = \frac{1}{4}e^{j\theta}$ and one at $z = 4e^{j\theta}$ along the dashed line), so this satisfies the zero-phase pattern.

For the inverse: $H(z)$ has zeros both inside and outside the unit circle. The inverse will therefore have poles both inside and outside the unit circle. To make that inverse stable we choose an ROC that includes

the unit circle and excludes those poles appropriately, which is possible but noncausal. Hence a stable noncausal inverse exists.

So, for (a): true. For (b): true (stable, but noncausal).

Case (d):

The zeros lie on the unit circle in conjugate-reciprocal fashion, so this is a zero-phase system.

For the inverse: the inverse would place poles exactly on the unit circle (at those zero locations). Poles on the unit circle mean the inverse cannot be stable.

So, for (a): true. For (b): false.

PROBLEM 5.61

The system function $H_{II}(z)$ represents a type II FIR generalized linear-phase system with impulse response $h_{II}[n]$. This system is cascaded with an LTI system whose system function is

$$G(z) = 1 - z^{-1}$$

to produce a third system with system function $H(z)$ and impulse response $h[n]$. Prove that the overall system is a generalized linear-phase system, and determine what type of linear-phase system it is.

A type II FIR generalized linear-phase system has odd order M (so length $M + 1$ is even) and an antisymmetric impulse response. Its frequency response can be written in the standard form

$$H_{II}(e^{j\omega}) = A_e(e^{j\omega})e^{-j\omega M/2}$$

where $A_e(e^{j\omega})$ is real, even, and periodic, and M is an odd integer.

Now consider

$$G(z) = 1 - z^{-1}.$$

Its frequency response is

$$G(e^{j\omega}) = 1 - e^{-j\omega} = e^{-j\omega/2}(e^{j\omega/2} - e^{-j\omega/2}) = e^{-j\omega/2} 2j \sin(\omega/2).$$

Thus we can write

$$G(e^{j\omega}) = A_G(e^{j\omega})e^{-j\omega/2}e^{-j\pi/2}$$

with

$$A_G(e^{j\omega}) = 2 \sin(\omega/2),$$

which is real and odd in ω , and the linear-phase term contributes a delay of $1/2$ sample plus a constant phase $-\pi/2$. This is exactly the form of a type IV FIR generalized linear-phase system.

The cascade has system function

$$H(e^{j\omega}) = H_{II}(e^{j\omega})G(e^{j\omega}) = A_e(e^{j\omega})e^{-j\omega M/2} \cdot A_G(e^{j\omega})e^{-j\omega/2}e^{-j\pi/2}.$$

After grouping the amplitude terms and the LP terms:

$$H(e^{j\omega}) = \underbrace{A_e(e^{j\omega})A_G(e^{j\omega})}_{A'(e^{j\omega})} e^{-j\omega(M/2+1/2)}e^{-j\pi/2}.$$

So

$$H(e^{j\omega}) = A'(e^{j\omega})e^{-j\omega(M+1)/2}e^{-j\pi/2}.$$

Since M is odd, $M + 1$ is even. Let $M' = M + 1$, so M' is even. Then

$$H(e^{j\omega}) = A'(e^{j\omega})e^{-j\omega M'/2}e^{-j\pi/2},$$

with $A'(e^{j\omega})$ real and odd (product of real even and real odd). A real, odd amplitude together with an even-length (M' even) linear-phase term corresponds to a type III FIR generalized linear-phase system: antisymmetric impulse response and odd length.

Therefore the overall cascade is still a generalized LP system, and specifically it is type III.

PROBLEM 4.43

All branches of the signal flow graphs in this problem have unity gain unless shown otherwise.

(a) Let the internal node signals of Figure P43-1 be as shown below:

$$\begin{aligned}w_1[n] &= x[n] + w_2[n-1] \\w_2[n] &= x[n] + w_3[n-1] \\w_3[n] &= 2w_1[n] + w_4[n-1] \\y[n] &= w_3[n] \\w_4[n] &= -y[n] - w_2[n].\end{aligned}$$

Take the z -transform of each equation:

$$\begin{aligned}W_1(z) &= X(z) + z^{-1}W_2(z) \\W_2(z) &= X(z) + z^{-1}W_3(z) \\W_3(z) &= 2W_1(z) + z^{-1}W_4(z) \\Y(z) &= W_3(z) \\W_4(z) &= -Y(z) - W_2(z).\end{aligned}$$

Substitute $W_4(z) = -Y(z) - W_2(z)$ into the third equation:

$$W_3(z) = 2W_1(z) + z^{-1}(-Y(z) - W_2(z)) = 2W_1(z) - z^{-1}Y(z) - z^{-1}W_2(z).$$

But $Y(z) = W_3(z)$, so

$$W_3(z) = 2W_1(z) - z^{-1}W_3(z) - z^{-1}W_2(z).$$

Bring the $W_3(z)$ terms together:

$$(1 + z^{-1})W_3(z) = 2W_1(z) - z^{-1}W_2(z).$$

Also from the first two equations,

$$W_1(z) = X(z) + z^{-1}W_2(z), \quad W_2(z) = X(z) + z^{-1}W_3(z).$$

Substitute these into the relation for $W_3(z)$:

$$(1 + z^{-1})W_3(z) = 2(X(z) + z^{-1}W_2(z)) - z^{-1}W_2(z) = 2X(z) + (2z^{-1} - z^{-1})W_2(z) = 2X(z) + z^{-1}W_2(z).$$

Now replace $W_2(z)$ by $X(z) + z^{-1}W_3(z)$:

$$(1 + z^{-1})W_3(z) = 2X(z) + z^{-1}(X(z) + z^{-1}W_3(z)) = 2X(z) + z^{-1}X(z) + z^{-2}W_3(z).$$

Collect the $W_3(z)$ terms:

$$(1 + z^{-1} - z^{-2})W_3(z) = (2 + z^{-1})X(z).$$

Since $Y(z) = W_3(z)$, the system function is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2 + z^{-1}}{1 + z^{-1} - z^{-2}}.$$

This is a second-order IIR (highest power in the denominator is z^{-2}), so an equivalent realization needs just 2 delays. Therefore, it's possible to implement the same input-output relationship using fewer delays, and the minimum number of delays is 2.

(b) For Figure P43-2 we can label the internal signals as

$$w_1[n], w_2[n], w_3[n], w_4[n], w_5[n]$$

from left to right, top branch being w_1, w_3, w_5 and bottom branch being w_2, w_4 . Then

$$\begin{aligned} w_1[n] &= x[n] + w_2[n-1] \\ w_2[n] &= x[n] + w_3[n-1] \\ w_3[n] &= 2w_1[n] \\ w_4[n] &= -w_3[n-1] - y[n] \\ w_5[n] &= w_3[n] + w_4[n-1] \\ y[n] &= w_5[n]. \end{aligned}$$

Take z -transforms:

$$\begin{aligned} W_1(z) &= X(z) + z^{-1}W_2(z) \\ W_2(z) &= X(z) + z^{-1}W_3(z) \\ W_3(z) &= 2W_1(z) \\ W_4(z) &= -z^{-1}W_3(z) - Y(z) \\ W_5(z) &= W_3(z) + z^{-1}W_4(z) \\ Y(z) &= W_5(z). \end{aligned}$$

Combining the last two equations yields

$$Y(z) = W_3(z) + z^{-1}W_4(z) = W_3(z) + z^{-1}(-z^{-1}W_3(z) - Y(z)) = W_3(z) - z^{-2}W_3(z) - z^{-1}Y(z).$$

Bring the $Y(z)$ terms together:

$$(1 + z^{-1})Y(z) = (1 - z^{-2})W_3(z).$$

But $W_3(z) = 2W_1(z)$ and $W_1(z) = X(z) + z^{-1}W_2(z)$; also $W_2(z) = X(z) + z^{-1}W_3(z)$. Substituting and simplifying yields

$$(1 + z^{-1})Y(z) = \frac{(1 - z^{-2})(2 + 2z^{-1})}{1 - 2z^{-2}}X(z),$$

so

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2(1 + z^{-1})(1 - z^{-1})}{1 - 2z^{-2}}.$$

Because this is not the same transfer function as in part (a), System B does not represent the same input-output relationship as System A.

PROBLEM SET 6, #1

The LTI systems $H_1(e^{j\omega})$ and $H_2(e^{j\omega})$ are generalized linear-phase systems. Which of the following systems are also generalized linear-phase systems?

$$\begin{aligned} G_1(e^{j\omega}) &= H_1(e^{j\omega}) + H_2(e^{j\omega}) \\ G_2(e^{j\omega}) &= H_1(e^{j\omega}) H_2(e^{j\omega}) \\ G_3(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_1(e^{j\theta}) H_2(e^{j(\omega-\theta)}) d\theta \end{aligned}$$

For a generalized linear-phase system, the frequency response has the form

$$H_i(e^{j\omega}) = A_i(\omega) e^{-j\omega n_i},$$

where $A_i(\omega)$ is real (either even or odd), and n_i is a real constant delay.

(1) For $G_1(e^{j\omega})$:

$$G_1(e^{j\omega}) = A_1(\omega)e^{-j\omega n_1} + A_2(\omega)e^{-j\omega n_2}.$$

This sum has a single linear phase only if $n_1 = n_2$ (the same delay) and both $A_1(\omega), A_2(\omega)$ have the same even/odd symmetry. If $n_1 \neq n_2$, the sum contains two distinct linear-phase terms and is not generally of

linear-phase form. Therefore G_1 is generalized linear-phase when the two systems have the same delay and type.

(2) For $G_2(e^{j\omega})$:

$$G_2(e^{j\omega}) = A_1(\omega)A_2(\omega)e^{-j\omega(n_1+n_2)}.$$

The product of two real even (or even/odd) amplitude terms is again real, and the phase is linear with slope $(n_1 + n_2)$. Hence G_2 is always a generalized linear-phase system, with delay $n_1 + n_2$.

(3) For $G_3(e^{j\omega})$:

$$G_3(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_1(e^{j\theta}) H_2(e^{j(\omega-\theta)}) d\theta$$

is a circular conv. of H_1 and H_2 in frequency. In the time domain, this corresponds to multiplication of the corresponding impulse responses:

$$g_3[n] = h_1[n] h_2[n].$$

The product of two symmetric (or antisymmetric) sequences is symmetric, but its phase is no longer purely linear in general because pointwise multiplication in time breaks the constant-delay structure in frequency. Therefore G_3 is not necessarily generalized linear-phase.